

CNM Quantum Technician Bootcamp

The Quantum Technician Bootcamp, part of CNM Ingenuity's Deep Dive immersive technology training program, prepares students for careers as technicians in the quantum, semiconductor, defense, advanced energy, and aerospace industries. The program is a full time (40 hours per week), 10-week hands-on learning environment held at CNM Quantum Learning Lab, a state-of-the-art training facility. The facility and the curriculum were jointly developed by CNM and Sandia National Labs.

The curriculum is 85% hands-on learning. The course leads to a stand-alone quantum technician certificate. While approximately 15% of students are also enrolled in a CNM academic, this is not a requirement. There are no math or science prerequisites.

Over the course of the 400 hours of training, students learn:

- Lab and Laser Safety, Lab Etiquette
- Optics and Photonics
- Ultra High Vacuum Systems
- Quantum Phenomenon
- Signal Generation, Amplification, Measurement
- Instrumentation and Data Collection
- Superconducting Qubit Calibration
- 3D Modeling in Solidworks
- Algebra, Trigonometry, Statistics
- Problem Solving, Documentation
- Introduction to Python Programming with exposure to Qiskit or CUDA-Q.
- 3D Printing, Laser Cutting, Introduction to Manual Milling
- IPC J-STD-001 REV J Soldering Certificate
- Fundamental of Lean Certificate

Class projects include the following. Note, all of the below projects are built by the students from individual components not premade assemblies.

- Optical Tweezers
- Michelson Interferometers
- Quantum Key Distribution
- Entangled Photon Pair (BBO Crystal) Quantum Optics Setup
- Rb Vapor Cell Spectroscopy
- Rb Magento-Optical Trap